C W-derived epithelial programs (K=8)

Five leading nodes from the validated exponential-decay selection (within node kind: τ=5, 95% cumulative mass).

P1 MHC-I antigen presentation / interferon-immune signaling

6 factors

	Δz
CD74	2.65
IGKC	2.52
1a_11b_dihydro_4_9_dimethylbenz_a_anthra_3_4_b_ oxirene	2.32
antigen processing and presentation of endogenous peptide antigen via MHC class I...	2.14
microsomal_epoxide_hydrolase	2.11

P2 RNA processing / glycosylation precursor metabolism

3 factors

	Δz
HMOX2	3.20
pseudouridine_5_phosphate	2.43
tRNA thio-modification (GO:0034227)	2.32
PMANM	2.26
GDP-mannose synthesis	2.05

P3 RNA splicing / chromatin and mitotic regulation

2 factors

	Δz
CLK1	4.73
G_5_pppR_RNA	3.94
histone H3-K36 methylation (GO:0010452)	3.91
DSG2	3.81
N3Tg	3.41

P4 Mitotic spindle assembly / chromosome segregation

28 factors

	Δz
ADM	0.36
ATF3	0.32
AURKB	0.30
TPX2	0.29
CXCL10	0.16

P5* Stem-like epithelial plasticity / matrix remodeling

singleton · factor 12

	Δz
RENBP	6.12
ACGAM2E	5.38
COL2A1	5.28
activated_methyl_group	3.18
positive regulation of voltage-gated calcium channel activity (GO:1901387)	2.57

P6 FGFR-associated developmental morphogenesis

3 factors

	Δz
FAM107A	3.99
FGFR3	3.72
salivary gland morphogenesis (GO:0007435)	3.21
S_2_aminobutanoate	3.11
HIBDm	3.02

P7 Folate-lipid metabolism / TNF-Notch survival signaling

2 factors

	Δz
STAG3	4.97
TNFRSF14	4.04
2_aminoethylphosphonate	3.92
folate transmembrane transport (GO:0098838)	3.42
MAR04872	3.34

P8 Mitochondrial redox / transsulfuration stress response

5 factors

	Δz
NDUFA13	2.93
selenocystathionine	2.38
intermediate filament bundle assembly (GO:0045110)	2.37
CYSTS	2.35
CD72	2.23

Node kind

gene · GO term · metabolite · cellchat entity · task · reaction

Display labels may be shortened for fit; exact identifiers and values are retained in the companion CSV.